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CS & DA

Warehousing

DPP: 3

Q1 Which of the following clustering algorithms is sensitive to outliers?
(A) K-means
(B) DBSCAN
(C) Hierarchical clustering
(D) PCA

Q2 In hierarchical clustering, which of the following methods involves merging clusters based on the closest pair of clusters?
(A) Ward's method
(B) Complete linkage
(C) Single linkage
(D) Average linkage

Q3 What is the time complexity of the K-means clustering algorithm?
(A) O(n)
(B) O(n log n)
(C) $O(n^2)$
(D) O(n*k)

Q4 Which of the following is NOT a level in the concept hierarchy of a data warehouse?
(A) Base level
(B) Aggregation level
(C) Detail level
(D) Composite level

Q5 Concept hierarchies in data mining are used for:
(A) Reducing dimensionality
(B) Handling missing data
(C) Organizing data into meaningful categories
(D) Evaluating classifier performance

Q6 Which of the following sampling methods ensures that each element in the population has an equal chance of being selected?
(A) Stratified sampling
(B) Convenience sampling
(C) Snowball sampling
(D) Simple random sampling

Q7 When is systematic sampling most appropriate?
(A) When the population is homogeneous
(B) When the population size is small
(C) When the sampling frame is not available
(D) When the sampling interval is known

Q8 Which of the following compression techniques is lossless?
(A) JPEG
(B) MP3
(C) GIF
(D) MPEG

Q9 Which step is NOT typically involved in lossless data compression?
(A) Quantization
(B) Run-length encoding
(C) Huffman coding
(D) Dictionary-based compression

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Answer Key

Q1	(B)
Q2	(C)
Q3	(D)
Q4	(D)
Q5	(C)
Q6	(D)
Q7	(D)
Q8	(C)
Q9	(A)

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Hints & Solutions

Q1 Text Solution:

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is sensitive to outliers because it defines clusters as dense regions of data points separated by regions of lower density, which can classify outliers as noise. K-means and hierarchical clustering are not as sensitive to outliers since they form clusters based on centroid or linkage distances respectively. PCA (Principal Component Analysis) is a dimensionality reduction technique, not a clustering algorithm.

Q2 Text Solution:

Single linkage clustering (also known as nearest neighbor clustering) merges clusters based on the smallest distance between points in different clusters. Ward's method, complete linkage, and average linkage use different criteria for merging clusters, such as minimizing variance (Ward's method) or using the maximum or average distances between clusters (complete and average linkage).

Q3 Text Solution:

In K-means clustering, where n is the number of data points and k is the number of clusters, the algorithm typically converges in O(n*k) time complexity. It iteratively assigns data points to clusters and updates cluster centroids until convergence.

Q4 Text Solution:

Concept hierarchies in data warehouses typically include base level (raw data), aggregation level (summarized data), and detail level (detailed data). The composite level is not a standard term in this context.

Q5 Text Solution:

Concept hierarchies organize data into meaningful categories or levels, which aids in understanding and analyzing data patterns. Reducing dimensionality, handling missing data, and evaluating classifier performance are tasks generally addressed by other techniques or methodologies in data mining.

Q6 Text Solution:

Simple random sampling ensures that each element in the population has an equal probability of being selected, without any bias towards certain characteristics. Stratified sampling divides the population into subgroups, convenience sampling selects subjects based on ease of access, and snowball sampling relies on referrals from initial subjects.

Q7 Text Solution:

Systematic sampling involves selecting every k-th element from a list after randomly selecting a starting point. It is efficient when the sampling interval (k) is known, making it appropriate for situations where the interval can be calculated or determined.

Q8 Text Solution:

GIF (Graphics Interchange Format) is a lossless compression format, meaning no data is lost during compression and decompression. JPEG, MP3, and MPEG are lossy compression formats, which sacrifice some data to achieve higher compression ratios.

Q9 Text Solution:

Quantization is a technique used in lossy compression to reduce the precision of data. In lossless compression, methods such as run-length encoding, Huffman coding, and dictionary-based compression are used to reduce the size of data without losing any information.

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